

Total No. of Questions : 10]

SEAT No. :

P6520

[Total No. of Pages : 2

[5870]-2001

T.E. (Computer Engineering)

DATA COMMUNICATION AND WIRELESS SENSOR NETWORKS

(2012 Pattern) (Semester - I) (310243)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume Suitable data if necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Explain Go Back N ARQ protocol with neat diagram. **[5]**

b) Justify the use of Wireless Sensor Network in Internet of Things. **[5]**

OR

Q2) a) How Character oriented approach of framing differs from Bit oriented approach of framing. **[5]**

b) List the properties a line code should have. Draw the following bit pattern in RZ, NRZ, AMI and Manchester code. **[5]**

Q3) a) Describe how Delta Modulation system works. What is meant by slope overload? **[6]**

b) Explain sensor network organization and tracking. **[4]**

OR

Q4) a) Explain how adaptive delta modulation improves systems tolerance to slope overload. **[6]**

b) Describe the working of Piconet. **[4]**

Q5) a) What is the main purpose of leach protocol? List and explain various approaches for the cluster selection in LEACH protocol. **[6]**

b) How hidden station and exposed station problem affects the communication in wireless network. Explain in brief. **[5]**

c) SMAC protocol is used for efficient energy utilization in WSN. Explain how? **[5]**

OR

P.T.O.

- Q6)** a) Explain techniques to handle Address and name management in wireless sensor networks. [6]
b) Differentiate between Contention-based protocols and Schedule-based protocols. [5]
c) With neat diagram, explain ZigBee technology. [5]
- Q7)** a) State with TRUE or FALSE with justification “SPIN uses attribute value pairs for data and queries”. [6]
b) What are different routing challenges and design issues in WSN? [5]
c) Write a short note on: [5]
i) Directed diffusion
ii) Rumor routing

OR

- Q8)** a) Explain Attribute based routing in WSN. [6]
b) Explain the Flooding technique and its drawbacks. [5]
c) Explain in detail data dissemination and gathering. [5]
- Q9)** a) Why localization is needed. Explain phases localization in detail. [8]
b) Write a note on joint routing and information aggregation in sensors. [5]
c) Explain information based sensor tracking with neat diagram. [5]

OR

- Q10)** a) Explain in detail the operating system design issues in WSN with reference to architecture, function etc. [8]
b) Explain the impact of anchor placement in WSN. [5]
c) How big data is handled in sensor network. [5]

